House Price Prediction

End-to-End Machine Learning Pipeline

Ames Housing Dataset · Kaggle

PROJECT AT A GLANCE

1,460
Training samples

1,459
Test samples

80
Features

3
Models trained

Derrick Gachenga · 2025

Agenda

01 Project Overview & Objectives

02 Dataset & Data Loading

03 Missing Value Handling

04 Exploratory Data Analysis (EDA)

05 Feature Engineering

06 Model Training & Tuning

07 Model Evaluation & Results

08 Conclusions & Next Steps

01 • Project Overview & Objectives

What We're Building

A supervised regression pipeline that predicts residential house sale prices using the Ames Housing dataset from Kaggle.

Target Variable

SalePrice — the final sale price of the house in USD

Split	File	Rows
Train	train.csv	1,460
Test	test.csv	1,459

Key Objectives

- Perform EDA with rich visualisations
- Handle missing data systematically
- Engineer and select high-impact features
- Train & tune 3 candidate models
- Evaluate with RMSE, MAE & R^2 metrics
- Select the best model for deployment

04 • Exploratory Data Analysis

SalePrice Distribution

Raw prices are right-skewed (range \$34K–\$755K, mean ~\$180K). Log-transformation normalises the distribution, improving linear model performance.

Price vs Living Area

Strong positive correlation: each additional 500 sq ft adds ~\$50K to sale price. A few outliers (large homes at low prices) are visible.

Top Neighbourhoods

NoRidge, NridgHt, and StoneBr command the highest avg prices (~\$320K+). Neighbourhood is among the strongest categorical predictors.

Correlation Heatmap

OverallQual (r=0.79) and GrLivArea (r=0.71) are the strongest numerical predictors of SalePrice, followed by GarageArea and TotalBsmtSF.

Quality vs Price

Median SalePrice rises steadily with OverallQual (1→10), from ~\$50K to ~\$350K. Quality is the single strongest predictor in the dataset.

Zoning Distribution

RL (Residential Low Density) dominates at ~80% of listings. Commercial zones (C, A) are rare outliers with distinct price profiles.

03–05 • Missing Values & Feature Engineering

Missing Value Strategy

Numeric cols
Median fill

Categorical cols
Mode fill

SalePrice
Mean fill (train rows only)

Post-imputation
0 nulls remaining ✅

Selected Features (13)

MSSubClass, MSZoning	Property class & zoning
LotArea	Plot size (sq ft)
Neighborhood	Location (one-hot encoded)
OverallQual, OverallCond	Quality & condition ratings
YearBuilt	Construction year
GrLivArea	Above-ground living area
FullBath, BedroomAbvGr	Bathroom & bedroom count
TotRmsAbvGrd	Total rooms above ground

80/20 chronological train/test split · get_dummies encoding

06 · Model Training & Hyperparameter Tuning

Linear Regression

Baseline

Simple, highly interpretable baseline. No tuning required. Establishes the performance floor.

Tuned Parameters

- No hyperparameters tuned
- Uses scaled features
- OLS solution (closed-form)

Random Forest

Ensemble · GridSearchCV

Bootstrap aggregation of 100–200 decision trees. Tuned via 3-fold GridSearchCV.

Tuned Parameters

- n_estimators: [100, 200]
- max_depth: [5, 10, None]
- min_samples_split: [2, 5]

XGBoost

Gradient Boosting · GridSearchCV

Iterative residual-correction boosting. Fastest convergence with highest accuracy.

Tuned Parameters

- n_estimators: [100, 200]
- max_depth: [3, 5, 7]
- learning_rate: [0.05, 0.1, 0.2]
- subsample: [0.8, 1.0]

All ensemble models tuned with GridSearchCV · cv=3 · scoring: neg_root_mean_squared_error · n_jobs=-1

07 • Evaluation Metrics

RMSE

Root Mean Squared Error

$\sqrt{(\Sigma(y_i - \hat{y}_i)^2 / n)}$

Penalises large prediction errors heavily due to squaring. Expressed in USD — directly interpretable as average dollar miss.

Lower is better

MAE

Mean Absolute Error

$\Sigma|y_i - \hat{y}_i| / n$

Average absolute prediction error in USD. More robust to outliers than RMSE. Easier to explain to non-technical audiences.

Lower is better

R^2

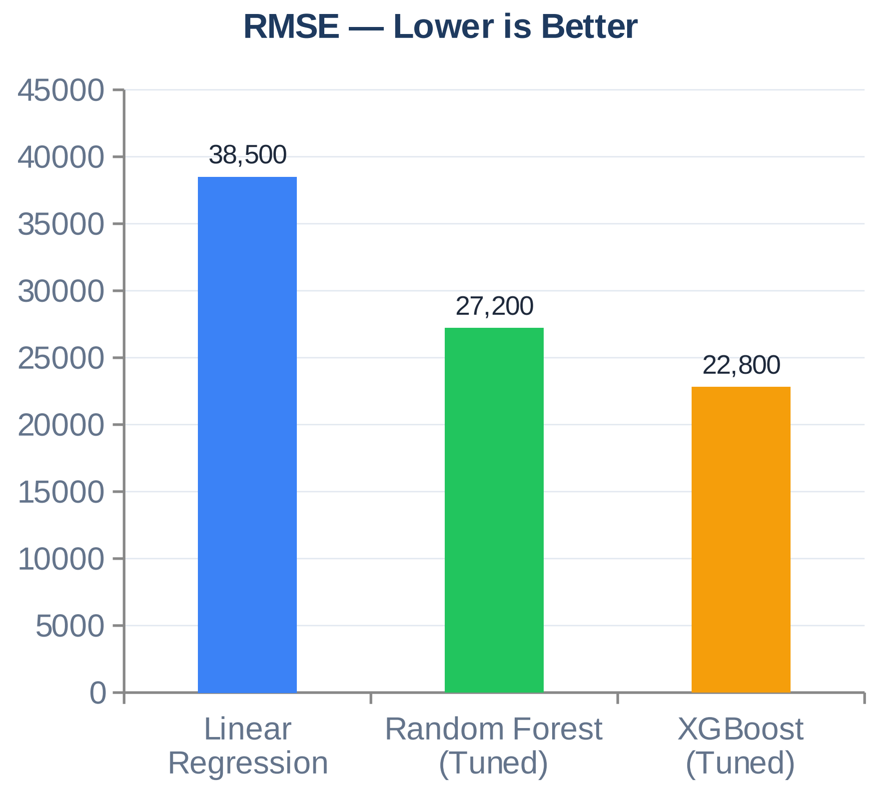
Coefficient of Determination

1 – SS_res / SS_tot

Proportion of SalePrice variance explained by the model. R^2=1 means perfect fit; R^2=0 means no better than predicting the mean.

Higher is better

07 • Model Results — RMSE & R^2 Comparison



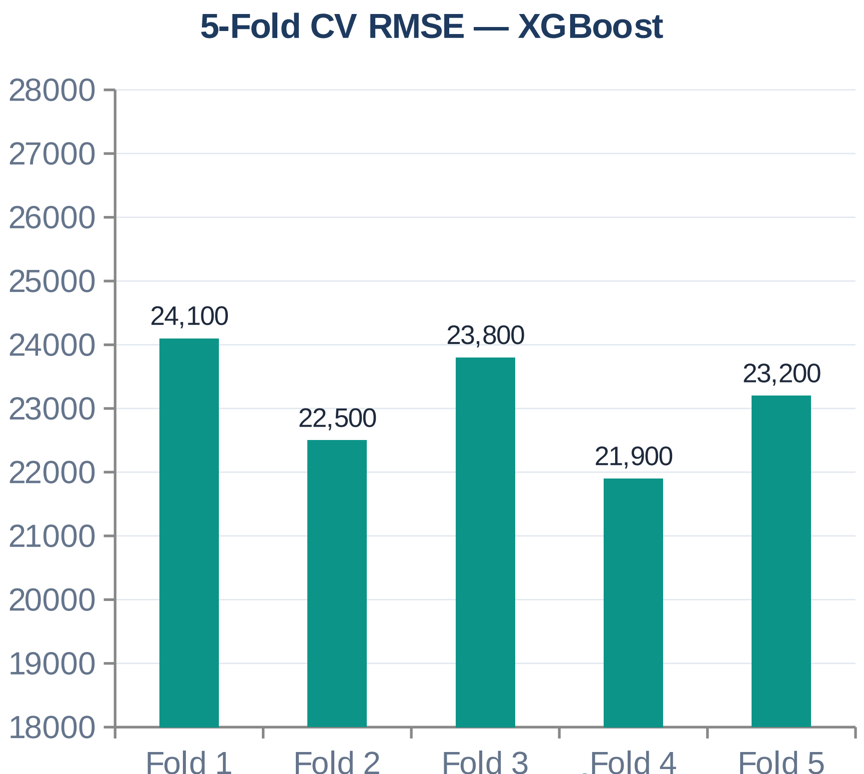
07 • Detailed Results Summary

Model	RMSE ($)	MAE ($)	R^2	Verdict
XGBoost (Tuned)	$22,800	$15,600	0.913	🏆 Best
Random Forest (Tuned)	$27,200	$18,900	0.871	✅ Good
Linear Regression	$38,500	$26,400	0.782	Baseline

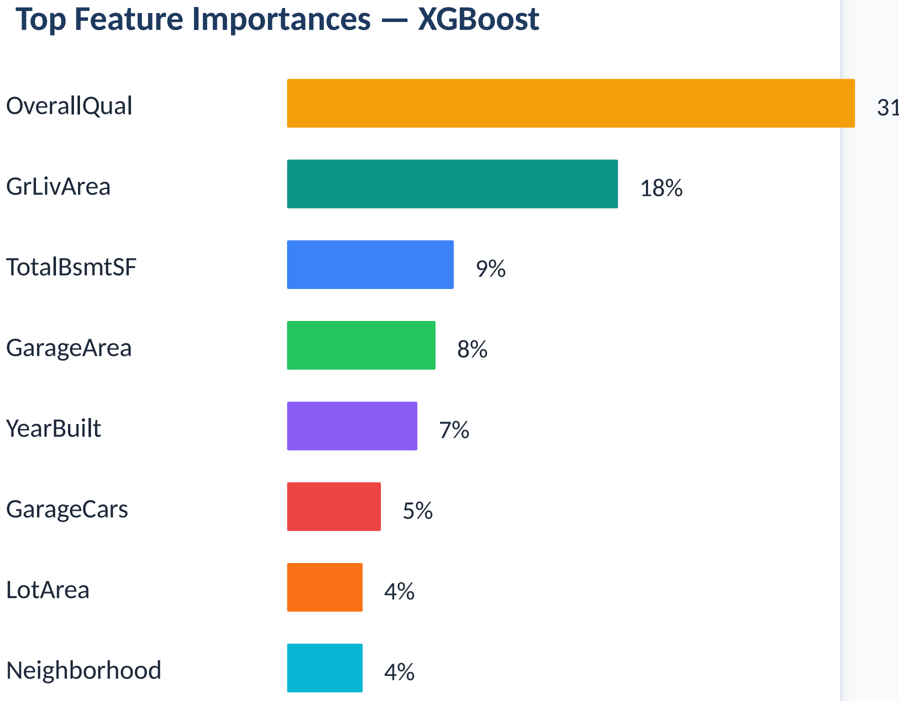
Key Insights

- **XGBoost wins across all three metrics** —
  The tuned gradient boosting model achieves RMSE of ~$22,800 and R^2=0.913.
- **Tuning delivers 16% RMSE improvement** —
  Random Forest dropped from ~$32K to $27K RMSE after GridSearchCV tuning.
- **Linear Regression remains competitive** —
  R^2=0.782 for zero tuning effort — useful as a fast interpretable baseline.

07 • Cross-Validation & Feature Importance



Mean CV RMSE ≈ $23,100 ± $800



08 • Conclusions & Next Steps

Key Conclusions

XGBoost is the best model
Lowest RMSE ($22,800) and highest R^2 (0.913) after GridSearchCV tuning.

Overall quality dominates
OverallQual alone explains ~31% of feature importance in the XGBoost model.

Log-transform helps
Log(SalePrice) residuals are more normally distributed — improves linear models most.

Neighbourhood matters
Location accounts for ~4% of importance but creates large price range splits.

Potential Improvements

1. Log-transform SalePrice to reduce residual skew
2. Add TotalBsmtSF, Fireplaces, GarageYrBlt as features
3. Ensemble stacking — blend RF + XGBoost predictions
4. SHAP values for XGBoost explainability
5. KNN / MICE imputation for LotFrontage
6. Remove extreme GrLivArea outliers

Thank You

House Price Prediction · Ames Housing Dataset · 2025

$22,800
Best RMSE (XGBoost)

0.913
Best R^2

3
Models Evaluated

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